Sequences and Limits

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First, an example

Before any abstraction, here is the ϵ–N game in miniature. Take the sequence $x_n = 1/n$ for $n = 1, 2, 3, \ldots$, and the candidate limit $L = 0$. We claim $x_n \to 0$.

Suppose a skeptic demands tolerance $\epsilon = 0.01$. We must exhibit an index N such that $|x_n - 0| < 0.01$ for every $n \geq N$. Choose $N = 100$: then for $n \geq 100$ we have $|x_n| = 1/n \leq 1/100 = 0.01$, so $|x_n| < 0.01$ holds for all $n > 100$ (and at $n = 100$ we get equality, easily fixed by $N = 101$). More generally, for any $\epsilon > 0$ pick $N = \lceil 1/\epsilon \rceil + 1$; the same calculation gives $|x_n| < \epsilon$ for all $n \geq N$. The whole rest of this lesson is about taking this single example seriously.

1 Setting

Throughout, (X, d) denotes a metric space (concept 09). A *sequence* in X is a function $x : \mathbb{N} \to X$, written $(x_n)_{n \in \mathbb{N}}$ or simply (x_n). We study what it means for such a sequence to settle down toward a point of X, both with and without naming the limit in advance.

2 Convergence

Definition 1 (convergence)**.** A sequence (x_n) in X *converges* to $L \in X$ if

$$\forall \epsilon > 0,\ \exists N \in \mathbb{N},\ \forall n \geq N,\ d(x_n, L) < \epsilon.$$

Read aloud: "for all epsilon greater than zero, there exists a natural number N such that, for all n greater than or equal to N, the distance from x_n to L is less than epsilon." We write $x_n \to L$ as $n \to \infty$, or $\lim_{n\to\infty} x_n = L$.

Remark 2. The integer N is allowed to depend on ϵ. The content of the definition is: *however small a tolerance ϵ you demand, the tail of the sequence eventually sits inside the open ball of radius ϵ around L, and stays there.*

Proposition 3 (uniqueness of limit)**.** *If $x_n \to L$ and $x_n \to L'$ in (X, d), then $L = L'$.*

Proof. Suppose $L \neq L'$ and let $\epsilon = d(L, L')/3 > 0$. Choose N_1 with $d(x_n, L) < \epsilon$ for $n \geq N_1$ and N_2 with $d(x_n, L') < \epsilon$ for $n \geq N_2$. For $n \geq \max(N_1, N_2)$ the triangle inequality gives $d(L, L') \leq d(L, x_n) + d(x_n, L') < 2\epsilon = \frac{2}{3} d(L, L')$, a contradiction. □

3 The Cauchy property and completeness

Definition 4 (Cauchy sequence)**.** A sequence (x_n) in X is *Cauchy* if

$$\forall \epsilon > 0,\ \exists N \in \mathbb{N},\ \forall m, n \geq N,\ d(x_m, x_n) < \epsilon.$$

Read aloud: "for all epsilon greater than zero, there exists a natural number N such that, for all m and n greater than or equal to N, the distance between x_m and x_n is less than epsilon."

The Cauchy condition speaks only of the sequence's internal behaviour: the terms eventually cluster within any tolerance, without naming a target.

Definition 5 (completeness)**.** A metric space (X, d) is *complete* if every Cauchy sequence in X converges to a point of X.

Example 6. $\mathbb{R}$ with the usual metric is complete; this is one of the defining features of the real line. By contrast, $\mathbb{Q}$ is not complete: the sequence of decimal truncations of $\sqrt{2}$ is Cauchy in $\mathbb{Q}$ but has no rational limit.

4 Convergent sequences are Cauchy

The central theorem of this lesson runs in one direction unconditionally: convergence always implies the Cauchy property. The converse requires completeness.

Theorem 7 (every convergent sequence is Cauchy)**.** *Let (X, d) be a metric space and let (x_n) be a sequence in X. If (x_n) converges, then (x_n) is Cauchy.*

Proof. Suppose $x_n \to L$ for some $L \in X$. Fix $\epsilon > 0$. We must produce $N \in \mathbb{N}$ such that $d(x_m, x_n) < \epsilon$ whenever $m, n \geq N$.

Apply the definition of convergence with tolerance $\epsilon/2 > 0$: there exists $N \in \mathbb{N}$ such that

$$n \geq N \quad \Longrightarrow \quad d(x_n, L) < \frac{\epsilon}{2}.$$

Read aloud: "n greater than or equal to N implies the distance from x_n to L is less than epsilon over two." Now let $m, n \geq N$. By the triangle inequality (from the metric axioms),

$$d(x_m, x_n) \leq d(x_m, L) + d(L, x_n) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Read aloud: "the distance from x_m to x_n is less than or equal to the distance from x_m to L plus the distance from L to x_n, which is less than epsilon over two plus epsilon over two, equals epsilon." Since $\epsilon > 0$ was arbitrary, the sequence (x_n) satisfies the Cauchy criterion. □

Remark 8. The $\epsilon/2$ split is the canonical move: we have two distances to control ($d(x_m, L)$ and $d(L, x_n)$), each can be made $< \epsilon/2$, and the triangle inequality combines them into a bound of ϵ. The same trick appears in proofs that continuous functions preserve Cauchy sequences, that uniform limits of continuous functions are continuous, and that completions exist.

5 Bolzano–Weierstrass

Theorem 9 (Bolzano–Weierstrass)**.** *Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.*

This is the prototypical compactness statement: boundedness plus closedness in $\mathbb{R}^d$ implies that every sequence has a convergent subsequence, with the limit lying in the set. The notion of *sequential compactness* (concept 11 and beyond) is the abstract generalisation.

6 Why the Cauchy criterion is useful

In applied work — and especially in numerical analysis and machine learning — we often cannot exhibit a limit in closed form, but we can monitor distances between successive iterates. Theorem 7 together with completeness of $\mathbb{R}^d$ tells us that *shrinking pairwise distances suffice*: if the iterates of an optimisation algorithm satisfy $\|x_m - x_n\| \to 0$ as $m, n \to \infty$, then they converge to a point of $\mathbb{R}^d$, regardless of whether we can name that point. This is the foundation of every fixed-point convergence proof, including Picard–Lindelöf (concept 30) and the construction of strong solutions of SDEs (concept 33).